

Lec_30

Analysis of Max_Min Algorithms:

- Time complexity ✓
 - Space Complexity ✓
 - No of Comparisons
⇓
of Elements
- ⇒ need

Enatt

1) TC

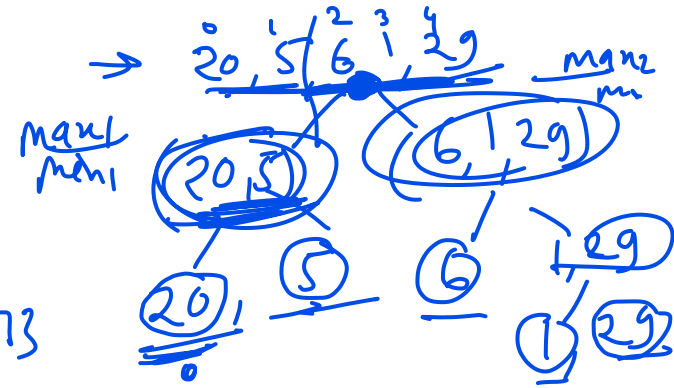
$DA(a, i, j)$ $a = \text{arr}$ $i, j \Rightarrow \text{index}$

Small

Base cases

```

if (i == j)
    return(a[i])
else
    max = min = a[i]
    if (a[i] > a[j])
        max = a[i]
    else
        max = a[j]
    min = a[i]
    if (a[i] < a[j])
        min = a[j]
    return(max, min)
    
```



Big
pr

```

else
    m = (i + j) / 2
    {
        max1, min1 = DA(a, i, m)
        max2, min2 = DA(a, m+1, j)
        if (max1 > max2)
            max = max1
        else
            max = max2
        if (min1 < min2)
            min = min1
        else
            min = min2
        return(max, min)
    }
    
```

TC if

$$T(n) = C + T(n/2) + T(n/2) + C$$

$$T(n) = 2T(n/2) + C$$

no of comparisons:-

$$C(n) = \begin{cases} 0 & n=1 \\ 1 & n=2 \\ 2C(n/2) + 2 & n > 2 \end{cases}$$

$$n^2 \Rightarrow T(n) = \underline{2T(n/2) + C}$$

~~Master's~~

$f(n)$ |
C

$$n \log_b a$$

$$n \log_2 2$$

$$TC = \cancel{\Theta(n)}$$

$$TC = \frac{O(n)}{\cancel{\Theta(n)}}$$

C | n

$\Theta(C)$ | $\Theta(n)$

3 no. of comparisons:-

base cond.

$C(n) = \begin{cases} 0, & n=1 \\ 1, & n=2 \\ 2C[n/2] + 2, & n > 2 \end{cases}$

Let $T(n)$

need exact ans

$$T(n) = 2T(n/2) + 2$$

use substitution method

$$T(n/2) = 2T(n/4) + 2$$

$$T(n/4) = 2T(n/8) + 2$$

$$2^3 + 2^2 + 2^1$$

$$T(n) = 2^2 T(n/4) + 2^2 + 2$$

$$T(n) = 2^3 T(n/8) + 2^3 + 2^2 + 2$$

$$T(n) = 2^3 T(n/8) + 2^3 + 2^2 + 2$$

$$T(n) = 2^n T(\frac{n}{2^n}) + 2^n + 2^{n-1} + 2^{n-2} + \dots + 2^1 + 2^0$$

if $\frac{n}{2^n} = 2$

$$2^n = \frac{n}{2}$$

$$T(n) = 2^n T(2) + 2^n + 2^{n-1} \dots - 21$$

$$T(n) = 2^n (1) + 2 + 2^2 + 2^3 \dots - 2^n$$

$$T(n) = 2^n + \frac{a(r^n - 1)}{r - 1}$$

$$T(n) = 2^n + 2 \left[\frac{2^n - 1}{2 - 1} \right]$$

$$T(n) = 2^n + 2 \cdot 2^n - 2$$

$$= \frac{n}{2} + \cancel{2} \cdot \frac{n}{\cancel{2}} - 2$$

$$= \frac{n}{2} + n - 2$$

$$= n \left(\frac{1}{2} + 1 \right) - 2$$

$$= 1.5n - 2$$

$$\frac{n}{2} = 2^n$$

no. of comparisons.

$$\boxed{1.5n - 2}$$

In DAE of minmax

each case

$$1.5n - 2$$

Conclusion of no. of comparisons
min_max

Without DAC

$$BC \rightarrow (n-1)$$

$$AC = 1.5(n-1)$$

$$WC = 2(n-1)$$

With DAC

$$1.5n - 2$$

each case

a $n = 54$, no. of comparisons to get min_max?

$$= \frac{3}{2} + \frac{54}{2} - 2 = 79 \dots \text{inference}$$

a $\Rightarrow$ $n-1 = 53$

TC

without DPC

$O(n)$

$O(n)$

$O(n)$

$O(1)$

SC

with DPC

$\Rightarrow$

$O(n)$

$O(n)$

$O(n)$

~~+~~

$O(\log n)$

Thanks

